

NETWORKS AND COMPLEXITY

Solution 23-1

*This is an example solution from the forthcoming book *Networks and Complexity*.
Find more exercises at <https://github.com/NC-Book/NCB>*

Ex 23.1: SIS on a three node chain [Lvl. 1]

In the chapter we introduced the interacting particle notation for the SIS model master equation. Use your intuition to formulate the Jacobian of the master equation for the SIS model on a chain of three nodes, (1)-(2)-(3).

Solution

We take the following into account:

- Node 1 can recover: $\mathbf{R} \otimes \mathbf{I} \otimes \mathbf{I}$
- Node 2 can recover: $\mathbf{I} \otimes \mathbf{R} \otimes \mathbf{I}$
- Node 3 can recover: $\mathbf{I} \otimes \mathbf{I} \otimes \mathbf{R}$
- Node 1 can infect node 2: $\mathbf{Q} \otimes \mathbf{P} \otimes \mathbf{I}$
- Node 2 can infect node 1: $\mathbf{P} \otimes \mathbf{Q} \otimes \mathbf{I}$
- Node 2 can infect node 3: $\mathbf{I} \otimes \mathbf{Q} \otimes \mathbf{P}$
- Node 3 can infect node 2: $\mathbf{I} \otimes \mathbf{P} \otimes \mathbf{Q}$

Hence, the Jacobian is

$$\mathbf{J} = \mathbf{R} \otimes \mathbf{I} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{R} \otimes \mathbf{I} + \mathbf{I} \otimes \mathbf{I} \otimes \mathbf{R} \quad (1)$$

$$+ \mathbf{Q} \otimes \mathbf{P} \otimes \mathbf{I} + \mathbf{P} \otimes \mathbf{Q} \otimes \mathbf{I} \quad (2)$$

$$+ \mathbf{I} \otimes \mathbf{Q} \otimes \mathbf{P} + \mathbf{I} \otimes \mathbf{P} \otimes \mathbf{Q} \quad (3)$$